# 3D-Printed Metamaterials

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# Outline

- Topic of the thesis.
- High permittivity and conductive material situations.
- Prospective topics in leaky-wave antennas.
- Prospective topics in reflectarray antennas.

# Topic of the thesis

- "Metamaterial" in the loosest sense, not limited to left handed materials.
- Focused only on topics utilizing dielectric, or lossy filaments
- Conventional 3DP techniques - FDM, SLA, SLS
- Specific topic is still to be decided.

# High Permittivity Materials

- High permittivity enables more applications, or more compact design with lower losses.
- FDM
  - Zetamix: $\varepsilon_r = 10$ with $\tan\delta 10^{-3}$, B2C
  - TPU: high permittivity, however $\tan\delta \approx 0.03$ 0.10.
- SLA
  - Rogers: RogersRadix – $\varepsilon_r = 2.8$.
  - 3Dresyns: $\varepsilon_r = 8 - 9$ with $\tan\delta < 0.04$, pricey
- SLS
  - Ceramics are difficult to print, more suitable to binder jetting,
  - Not considered.

# Conductive Materials

- Carbon-loaded filaments possess low conductivity and behave primarily as lossy dielectrics.
  - These low-conductivity materials excel as monolithic electromagnetic absorbers.
  - Proto-pasta has PLA loaded with carbon black.
- Copper-loaded composites achieve significantly higher bulk conductivities.
  - High-conductivity materials effectively mimic perfect conductors at microwave frequencies, enabling single-step metamaterial fabrication.
  - They are pricey, not considered.

# LWA: Monolithic Multi-Material Matched Terminations

- Unradiated residual power at the aperture end reflects and generates high sidelobes[1].
- Multi-head fused deposition modeling permits co-printing a structural dielectric and a lossy composite absorber.
- Continuous spatial impedance tapering suppresses back-reflection and ensures reflectionless energy transfer.
- Static dissipative polylactic acid or conductive polyurethane can be utilized for distributed attenuation.

[1] Francesco Monticone et al. "Leaky-Wave Theory, Techniques, and Applications: From Microwaves to Visible Frequencies". In: *Proceedings of the IEEE* 103.5 (2015), pp. 793–821.

# LWA: Continuous Sinusoidal Reactance Surfaces

- Most basic LWAs discretize periodic modulation into square pulses (blocks of different $\varepsilon_r$ materials, air), exciting unwanted higher-order space harmonics.
- Commonly a more complex unit cell design or heigh modulation in $z$-axis is used to solve this.
- Different Approach: Varying infill or using foaming PLA
- More complex solutions then already existing, but final structure is a slab on the outside

# LWA: Chirped Conformal Pre-Distorted Geometries

- Mounting a periodic leaky-wave antenna onto a curved surface introduces severe non-linear phase errors.
- Additive manufacturing enables phase correction by continuously modifying the internal geometric fill fraction[2].
- Inverse-design methodology progressively chirps the unit-cell periodicity along the curved propagation axis.
- The conformal structure directly synthesizes a collimated high-gain plane wave from a highly curved physical surface.

[2] Patrick Bartley et al. "Conformal Design for 3D-Printable Leaky-Wave Antennas". In: *2025 International Symposium on Antennas and Propagation (ISAP)*. 2025, pp. 1–2.

# Reflectarrays: Graded-Index via Infill Density Control

- Conventional dielectric reflectarrays rely on geometric height variations that introduce shadowing and diffraction.
- Spatially modulating the internal fill fraction tunes the local effective permittivity while maintaining a flat external profile.
- This continuous tuning enables completely flush-mounted reflectarrays for conformal aerospace applications.
- The process utilizes high-permittivity ceramic filaments and requires programmatic slicer automation[3].

[3] Michele Beccaria et al. "Ka-Band Reflectarray with Cylindrical Dielectric Unit Cells: Optimized Additive Manufacturing and High-Permittivity Material Characterization". In: *Sensors* 25.17 (2025).

# Reflectarrays: Sub-THz Structures for 6G

- Printed circuit board fabrication tolerances and conductor losses become prohibitive above 100 GHz.
- Spatial feeding in reflectarrays entirely eliminates the lossy corporate feed network.
- All-dielectric unit cells mitigate high-frequency conductor losses.
- This direction establishes the practical upper frequency limits of additive manufacturing for telecommunication systems.

# Reflectarrays: Conformal Curved Structures

- Standard planar phase synthesis produces severe beam defocusing when applied to arbitrarily curved surfaces.
- Phase distributions must be recalculated – to account for local surface normal rotation.
  - Metasurface with high phase stability is required – range if incidence angle is larger then in planar arrays
- Additive manufacturing permits direct fabrication onto pre-shaped (with significant slicer modification) substrates or as free-standing curved structures.
- The large design space necessitates programmatic computer-aided design and surrogate-based optimization algorithms.

# Reflectarrays: Integrated Dielectric Rod Feed

- Conventional commercial horn antennas introduce alignment uncertainty and prevent complete system miniaturization.
- A dielectric rod antenna can be printed simultaneously as a vertical protrusion directly from the array surface.
- This monolithic fabrication eliminates alignment variables and reduces overall component costs.
- The fixed focal length demands extreme simulation accuracy prior to the printing process.

# Bibliography


[1] Patrick Bartley et al. "Conformal Design for 3D-Printable Leaky-Wave Antennas". In: *2025 International Symposium on Antennas and Propagation (ISAP)*. 2025, pp. 1–2.

[2] Michele Beccaria et al. "Ka-Band Reflectarray with Cylindrical Dielectric Unit Cells: Optimized Additive Manufacturing and High-Permittivity Material Characterization". In: *Sensors* 25.17 (2025).

[3] Francesco Monticone and Andrea Alù. "Leaky-Wave Theory, Techniques, and Applications: From Microwaves to Visible Frequencies". In: *Proceedings of the IEEE* 103.5 (2015), pp. 793–821.